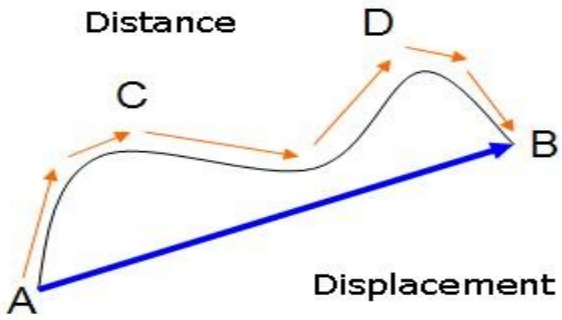
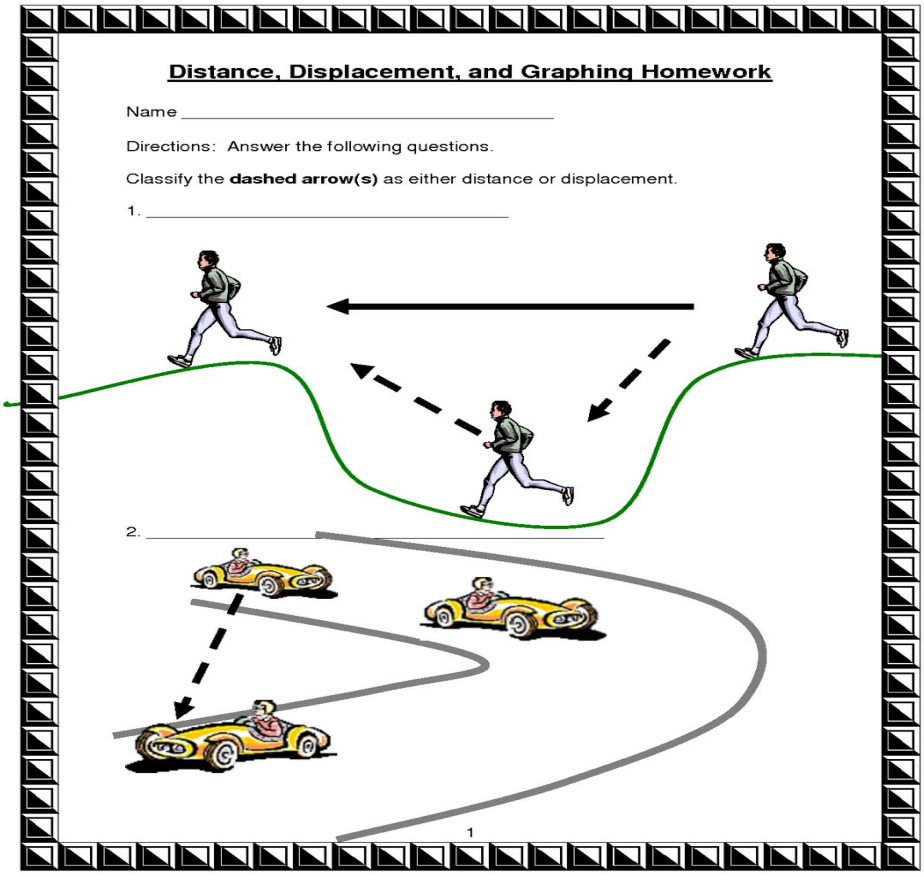
Name:

Distance and Displacement Practice

Distance = "how much ground an object has covered" during its motion. The total distance you had to cover to get somewhere.
Displacement=the straight line distance from the starting point to the finishing point.



In the diagram above the movement from A to C to D to B is the distance. The distance between A and B is the Displacement.



Calculate the *distance* and *displacement* in each of the following situations

1. Rochelle runs 2km south, then turns around and runs 3km north.
   What distance did she run?
   What is her displacement?

2. Paul runs exactly 2 laps around a 400 meter track.
   What distance did he run?
   What is his displacement?

3. Thalia spins around 5 times without moving any direction.
   What distance did she move?
   What is her displacement?

4. Tom walks two miles north from his door to the park, then returns home to his door.
   What distance did he move?
   What is his displacement?

5. Kiri ran 8 blocks north, and then 2 blocks south back toward her starting point.
   What distance did she move?
   What is her displacement?

6. Mohammad dove in at the south end of the swimming pool then swam 3 complete laps of the 50 meter pool. (1 lap is one length of the pool.)
   What distance did he swim?
   What is his displacement?

7. Mihi walks east for 3 km, stops for a break, and then runs the same direction for 4 km before she stops.
   What distance did she move?
   What is her displacement?

8. Cameron rides his bicycle 20 km north, turns around, and then rides the bicycle 15 km back toward his starting point.
   What distance did he move?
   What is his displacement?